Assignment_unit-1(oop)

1. What are the key features of Object-Oriented Programming?
2. What are the four fundamental principles of Object-Oriented Programming (OOP)?
3. Explain the concept of **platform independence** in Java.
4. What is **bytecode** in Java, and how does it help in achieving platform independence?
5. How does the Java compiler convert source code into bytecode?
6. What role does the **Java Virtual Machine (JVM)** play in running Java bytecode?
7. Explain the difference between **Java source code** and **bytecode**.
8. How do **objects** and **classes** relate to each other in OOP?
9. What is the **Java Development Kit (JDK)**, and what are its components?
10. Explain the difference between the **JDK** and the **Java Runtime Environment (JRE)**.
11. What is the **Java Virtual Machine (JVM)**, and how does it relate to Java execution?
12. What is the role of the **JVM** in **memory management** (e.g., garbage collection)?
13. What is a **class** in Java? Explain how to declare and define a class in Java.
14. What are **methods** in Java, and how are they defined and invoked?
15. Explain the role of operators in Java. List out all operators in Java and Explain each with an example.